

Sukruta Prakash Midigeshi

[Google Scholar](#)
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Education

Indian Institute of Technology, Gandhinagar
B.Tech in Computer Science and Engineering

Nov 2020 – June 2024
CGPA: 9.07/10

Publications and Preprints

Achieving Limited Adaptivity for Multinomial Logistic Bandits

SP Midigeshi, T Goyal, G Sinha | RLC'25

Characterizing deep research: A benchmark and formal definition

A Java, A Khandelwal*, SP Midigeshi*, A Halfaker, A Deshpande, N Goyal, A Gupta, N Natarajan, A Sharma | ICLR'26, SEA@NeurIPS '25*

Plan-RAG: Planning-guided Retrieval Augmented Generation

P Verma, SP Midigeshi, G Sinha, A Solin, N Natarajan, A Sharma | Workshop on Reasoning and Planning for Large Language Models@ICLR'25

Batched Contextual Slate Bandits with GLM Rewards

T Goyal, SP Midigeshi*, G Sinha*

Under Review

Work Experience

Research Fellow, Microsoft Research

Advisor- Dr.Gaurav Sinha

July 2024 - Present

- Developed batched algorithms for logistic and multinomial contextual bandits under limited adaptivity, with proven theoretical guarantees and strong empirical performance. Currently extending these results to large action spaces in contextual GLMs.
- Built a SOTA RAG pipeline (Plan-RAG) outperforming Self-RAG and ReAct; introduced LiveDR-Bench and formalized DeepResearch. Now working on improving the expressivity of reasoning models.

Software Intern, D.E.Shaw

Advisor - Mr.Devender Lega

May 2023 - June 2023

- Created evaluation metrics for LLMs across structured and unstructured datasets; analyzed models such as BERT, RoBERTa, and other open-source variants.
- Developed a recursive AI architecture built on TAPAS for question answering over custom structured data, addressing the lack of suitable training datasets.

Software Intern, eLitmus

Advisor - Mr.Shireesh Jayashetty

January 2022 - June 2022

- Worked on a face-recognition system and a source-code plagiarism detector extending standard tools like MOSS and JPlag.
- Built a cross-language plagiarism checker using Lark-based parse trees and an N-gram similarity model, enabling detection despite renaming, reordering, and syntactic differences.

Software Intern, goGlocal

Advisor - Mr.Shubham Kedia

June 2022 - July 2022

- Automated HS (Harmonized System) code generation from user images and descriptions using web-scraping pipelines and ML architectures.

Featured Projects and Collaborations

Approximating Neural Tangent Kernels using LSH

w/ Prof. Anirban Dasgupta

August 2023 - November 2023

- Explored methodologies for approximating Neural Tangent Kernels (NTKs), including the use of Locality-Sensitive Hashing (LSH) for kernel ridge regression.
- Extended LSH-based techniques beyond shift-invariant kernels to NTKs, establishing theoretical bounds and conducting experiments to identify the most effective approximation methods.

Envy-Free Budget-Feasible Allocations

w/ Prof. Neeldhara Misra

January 2022 - April 2022

- Studied algorithms producing EF1 and α -EF1 allocations in fair division, and designed an exact EF1 algorithm for the 2-agent identical-valuations setting with polynomial complexity.

- Analyzed the time complexity of key envy-free allocation algorithms, including ExistsUMandEF1, which determines whether an EF1 allocation can exist within a UM allocation.

Study on properties of undirected power graphs and their partitions.

August 2023 - November 2023

w/ Prof. Bireswar Das

- Studied properties of power graphs of finite abelian groups, including triangle partitions and clustering coefficients.
- Developed an algorithm to partition subsets of abelian groups into triangular segments, proposed an efficient formula for the global clustering coefficient of their power graphs, and designed a specialized algorithm for cyclic abelian groups.

Teaching Experience

Teaching Assistant, Machine Learning

January 2024 - May 2024

Prof. Nipun Batra

- Teaching Assistant for a course with ~200 students: created programming assignments, designed tutorial problem sets, and mentored students by addressing routine doubts.
- Assisted in grading exams and assessments, including conducting viva-based evaluation and helping manage large-scale course logistics.

ADH (Academic Discussion Hours) Mentor, Theory of Computing

January 2024 – May 2024

Prof. Balagopal Komarath

- Mentored ~100 students during Academic Discussion Hours, helping clarify doubts and explaining course concepts in Theory of Computing.
- Created sample questions for exams and provided personalized guidance and problem-solving support to strengthen students' understanding.

Teaching Assistant, Discrete Mathematics

January 2023 - April 2023

Prof. Bireswar Das

- Helped in the organization and structuring of the course - Discrete Mathematics offered with ~60 students
- Facilitated engaging discussions, assisted with problem-solving sessions, and provided guidance to students.

Achievements and Awards

- Received the Dean's List for excellence in academics for the semesters 2, 5, 6, 7 and 8.
- Earned the highest grade (11/10) in Natural Language Processing course for exceptional performance in a class of ~100 students.
- Secured 99.69 percentile in JEE Mains among 1.5 million applicants.
- Ranked in the top 1.6% among 200,000 students who qualified for JEE Advanced

Relevant Courses

- Discrete Mathematics [A], Machine Learning [A], Compilers [A-], Complexity Theory [A], Advanced Algorithms [A-], Data Science [A], Natural Language Processing [A+], Theory of Computing [A-], Combinatorics with Applications in Computer Science [A], Deep Learning [A]

Leadership and Extracurricular Achievements

- **Design Coordinator, Blithchron '21 (Cultural Fest - IITGN):** Helped lead the design team of Blithchron, the cultural fest of IIT GN. Coordinated between all the departments of the organising committee and managed the team.
- **Second Seed, Inter-IIT'22 Squash Team:** Represented IIT Gandhinagar as a member of the squash team at the prestigious Inter-IIT Sports Meet 2022.
- **Founding Member, WiT:** Founded and led an initiative to increase female participation in Computer Science research and industry.
- **PAL Mentor:** Conducted regular mentorship sessions to support students in improving their English language proficiency